MILITARY SPECIFICATION

LITHIUM CHLORIDE
(For Use in Ammunition)

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and packaging for the manufacture of one grade of lithium chloride for use in Thionyl Chloride ($SOCl_2$), reserve type batteries (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipping and Storage
MIL-STD-1218 - ACS Chemicals

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM-E70 PH of Agueous Solutions with Glass Electrode, Test for
ASTM-E276 Particle Size or Screen Analysis at No. 4 Sieve and Finer for Metal Bearing Ores and Related Material
ASTM-E203 Water Using Karl Fischer Reagent, Test for
ASTM-E300 Sampling Industrial Chemicals
ASTM-C562 Moisture in Graphite
ASTM-D2905 Statements on Number of Specimens required to Determine the Average Quality of A Textile Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013).

(Technical society and technical association specifications and standards are generally available for reference from libaries. They are also distributed among technical groups and using federal agencies)

3. REQUIREMENTS

3.1 Material. The material shall be lithium chloride having D values which conform to the D values as listed in the X-ray Powder Data Files of ASTM when tested in accordance with 4.6.5 and shall also conform to the following requirements:

TABLE 1
Requirements for Lithium Chloride

Property	Requirement	Applicable Paragraph
Purity, Minimum, %	98.0	4.6.1
Moisture, Maximum, (Max) %	1.0	4.6.2
pH	6.5 - 9.0	4.6.3
Granulation		4.6.4
Retained on US #20 mesh Max., %	10	
Iron, Max. ppm	200	4.6.6

3.2 Workmanship. The composition shall be free of dirt, chips, and other foreign matter.

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3.3 First Article Inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First Article Inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 200 grams of lithium chloride obtained by sampling as described in 4.5.1 (smaller weight samples may be taken that are consistent with the requirements of the test). The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production (see 6.3).

4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations for tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of lithium chloride produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of lithium chloride that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (see MIL-A-48078). The product shall be submitted for inspection in accordance with MIL-STD-105.

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4.5 Testing (see 6.4). Precaution: This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Two (2) individual 200 grams samples from each batch or lot of the lithium chloride shall be sampled as specified in ASTM E300, section 25. The sample shall be tested independently in accordance with the applicable methods given in 4.6. If either sample fails to meet any of the test requirements, the lot shall be rejected. The samples shall be subjected to the tests listed below:

TABLE II

Inspection/Requirement	Defect Classification
Material (see 3.1)	Major
Properties (see 3.1)	Major

4.4.3 Inspection equipment. The contractor shall use commercial inspection equipment for all tests and examinations specified in para 4.6. He is also charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use.

4.6 Test methods and procedures (see 6.2 and 6.4). The tests to determine compliance with the requirements given in Table I shall be performed on the lithium chloride samples obtained in 4.5.1. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be reagent grade or ACS Grade in accordance with MIL-STD-1218.

4.6.1 Purity. Carefully transfer an accurately weighed portion of the sample (approximately 10 grams of the lithium chloride) to a tared weighing bottle. The weighing shall be made to the .1 mg. Care should be exercised against contamination with moisture. Record this weight. Dissolve this sample in 100 ml of distilled water and transfer quantitatively to a 500 ml volumetric flask and make up to volume with distilled water. Transfer a 10 ml aliquot to a 250 ml Erlenmeyer flask and add 100 ml distilled water plus 5 ml concentrated nitric acid. Add slowly while shaking, 50 ml of

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N/10 silver nitrate solution. Add 5 ml of 10% ferric nitrate solution containing 5% nitric acid. Then add 15-20 drops nitrobenzene and shake vigorously, titrate the excess silver nitrate with N/10 potassium thiocyanate to first permanent reddish brown color. (Note: Silver nitrate should be restandardized with primary standard sodium chloride prior to use in the same manner as the sample is analyzed).

$$\% \text{ Lithium Chloride} - \frac{[(A \times N_1) - (B \times N_2)]}{W} \quad 4.240$$

Where:

A = ml of Silver Nitrate used
N_1 = Normality of Silver Nitrate Solution
B = ml of Potassium Thiocyanate used
N_2 = Normality of Potassium Thiocyanate Solution
W = Grams of sample corrected for moisture

4.6.2 Moisture.

4.6.2.1 Karl Fischer method. The moisture shall be determined in accordance with ASTM-E203.

4.6.2.2 Oven method. The moisture shall be determined in accorcance with ASTM-C562.

4.6.3 pH. The pH shall be determined in accordance with ASTM-E70 except a 2% solution shall be used.

4.6.4 Granulation. The granulation shall be determined in accordance with ASTM-E389 except a No. 20 sieve shall be used.

4.6.5 Material x-ray diffraction pattern. Determine the X-ray diffraction pattern of the sample using any standard powder diffraction method appearing in a standard or reference textbook (see 6.2).

4.6.6 Determination of iron. by Atomic Absorption spectrophotometer.

4.6.6.1 Standards.

Dissolve 100 mg of iron wire in a 1 to 1 nitric acid. Add to a one liter volumetric flask and dilute to mark with distilled water. Prepare standards of 2 ppm, 4 ppm, 6 ppm and 8 ppm in 50 ml volumetric flask and dilute to mark.

4.6.6.2 Procedure. Dissolve 5 grams of sample in 1 ml of hydrochloric acid and 2 ml of nitric acid in a 25 ml beaker. Transfer the solution to a 100 ml volumetric flask and dilute to the mark with distilled water.

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Adjust the Atomic absorption spectrophotometer to the manufacturers recommendations using an air/acetylcne flame with a deuterium background corrector. Measure and record the absorbance reading of the sample and standard.

Calculations:

$$\text{ppm of iron} = \frac{\text{ppm of the sample } 100}{\text{(wt of sample)}}$$

5. PACKAGING

5.1 Packing.

5.1.1 Drums. Unless otherwise specified in the contract or purchase order, the material shall be packed in non-returnable fiber drums in accordance with Specification PPP-D-723, (Type III Grade A). The drums shall be finished with a nominal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure.

5.1.2 Level C. The material shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipment or with National Motor Freight Rules and Container Specifications for truck shipments as applicable.

5.2 Marking. Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

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6.3 Lithium chloride which has been found satisfactory for use in lithium batteries is manufactured by Foote Mineral Company, Exton PA 19341 and MCB Manufacturing Chemists, 2909 Highland Avenue, Norwood, Ohio 45212. Lithium chloride from other sources should be evaluated by ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801 before being approved as an equivalent.

6.4 The tests described in this specification should be performed in a inert atmosphere glove box, or in a dry room with relative humidity not to exceed 5% at ambient conditions.

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